

LAB 7

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#define BUFFER_SIZE 5
#define NUM_PRODUCERS 2
#define NUM_CONSUMERS 2
#define NUM_ITEMS 10
int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;
sem_t empty;
sem_t full;
pthread_mutex_t mutex;
void* producer(void* param) {
    int producer_id = *((int*)param);
    for (int i = 0; i < NUM_ITEMS; i++) {
        int item = rand() % 100;
        sem_wait(&empty);
        pthread_mutex_lock(&mutex);
        buffer[in] = item;
        printf("Producer %d: Inserted item %d at index %d\n", producer_id, item, in);
        in = (in + 1) % BUFFER_SIZE;
        pthread_mutex_unlock(&mutex);
        sem_post(&full);
        sleep(1);
    }
    return NULL;
}
```

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}

void* consumer(void* param) {
    int consumer_id = *((int*)param);
    for (int i = 0; i < NUM_ITEMS; i++) {
        sem_wait(&full);
        pthread_mutex_lock(&mutex);
        int item = buffer[out];
        printf("Consumer %d: Removed item %d from index %d\n", consumer_id, item, out);
        out = (out + 1) % BUFFER_SIZE;
        pthread_mutex_unlock(&mutex);
        sem_post(&empty);
        sleep(2);
    }
    return NULL;
}

int main() {
    pthread_t producers[NUM_PRODUCERS];
    pthread_t consumers[NUM_CONSUMERS];
    int prod_ids[NUM_PRODUCERS];
    int cons_ids[NUM_CONSUMERS];
    pthread_mutex_init(&mutex, NULL);
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    for (int i = 0; i < NUM_PRODUCERS; i++) {
        prod_ids[i] = i + 1;
        pthread_create(&producers[i], NULL, producer, &prod_ids[i]);
    }
    for (int i = 0; i < NUM_CONSUMERS; i++) {
        cons_ids[i] = i + 1;
        pthread_create(&consumers[i], NULL, consumer, &cons_ids[i]);
    }
}

```

```
}  
for (int i = 0; i < NUM_PRODUCERS; i++) {  
    pthread_join(producers[i], NULL);  
}  
for (int i = 0; i < NUM_CONSUMERS; i++) {  
    pthread_join(consumers[i], NULL);  
}  
pthread_mutex_destroy(&mutex);  
sem_destroy(&empty);  
sem_destroy(&full);  
  
printf("All threads have finished execution.\n");  
return 0;  
}
```

OUTPUT

```
C:\Users\student\Downloads\ x + v
Producer 1: Inserted item 41 at index 0
Consumer 2: Removed item 41 from index 0
Producer 2: Inserted item 41 at index 1
Consumer 1: Removed item 41 from index 1
Producer 2: Inserted item 67 at index 2
Producer 1: Inserted item 67 at index 3
Consumer 1: Removed item 67 from index 2
Producer 1: Inserted item 34 at index 4
Producer 2: Inserted item 34 at index 0
Consumer 2: Removed item 67 from index 3
Producer 2: Inserted item 0 at index 1
Producer 1: Inserted item 0 at index 2
Consumer 2: Removed item 34 from index 4
Consumer 1: Removed item 34 from index 0
Producer 2: Inserted item 69 at index 3
Producer 1: Inserted item 69 at index 4
Producer 1: Inserted item 24 at index 0
Consumer 1: Removed item 0 from index 1
Consumer 2: Removed item 0 from index 2
Producer 2: Inserted item 24 at index 1
Producer 1: Inserted item 78 at index 2
Consumer 2: Removed item 69 from index 3
Consumer 1: Removed item 69 from index 4
Producer 2: Inserted item 78 at index 3
Producer 1: Inserted item 58 at index 4
Consumer 1: Removed item 24 from index 0
Consumer 2: Removed item 24 from index 1
Producer 1: Inserted item 62 at index 0
Producer 2: Inserted item 58 at index 1
Consumer 2: Removed item 78 from index 2
Consumer 1: Removed item 78 from index 3
Producer 2: Inserted item 62 at index 2
Producer 1: Inserted item 64 at index 3
Consumer 2: Removed item 58 from index 4
Consumer 1: Removed item 62 from index 0
Producer 2: Inserted item 64 at index 4
Consumer 1: Removed item 58 from index 1
Consumer 2: Removed item 62 from index 2
Consumer 2: Removed item 64 from index 3
Consumer 1: Removed item 64 from index 4
All threads have finished execution.

Process returned 0 (0x0)   execution time : 20.113 s
Press any key to continue.
|
```